

Historical Simulation (HS)

Portfolio VaR Engine — Model Documentation

Risk Modelling — Trade Republic

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1 Overview

Historical Simulation (HS) is the simplest and most transparent VaR methodology. It makes no assumptions about the return distribution: VaR is estimated directly as an empirical quantile of the observed portfolio return series. The method is therefore fully non-parametric and requires no model calibration.

Engine key: 'hs'

Source: `risk_analytics/risk_engines.py`, lines 228–250

2 Methodology

2.1 Portfolio Return Construction

Given a matrix of asset returns $\mathbf{R} \in \mathbb{R}^{T \times N}$ and a weight vector $\mathbf{w} \in \mathbb{R}^N$, the portfolio return series is:

$$r_t^{pf} = \sum_{i=1}^N w_i r_{i,t}, \quad t = 1, \dots, T \quad (1)$$

Only the most recent T observations are used (controlled by the `window` parameter, default $T = 250$).

2.2 Value-at-Risk

The 1-day VaR at tail probability α is the negative of the α -quantile of the empirical distribution:

$$\widehat{\text{VaR}}_\alpha = -Q_\alpha\left(\{r_t^{pf}\}_{t=1}^T\right) \quad (2)$$

where $Q_\alpha(\cdot)$ denotes the empirical α -th quantile. A positive value indicates a loss. Default quantiles: $\alpha \in \{0.05, 0.01, 0.001\}$.

2.3 Conditional Value-at-Risk (Expected Shortfall)

CVaR is the expected loss conditional on the loss exceeding the VaR threshold:

$$\widehat{\text{CVaR}}_\alpha = -\frac{1}{|\mathcal{T}_\alpha|} \sum_{t \in \mathcal{T}_\alpha} r_t^{pf} \quad (3)$$

where $\mathcal{T}_\alpha = \{t : r_t^{pf} \leq Q_\alpha\}$ is the set of dates where the portfolio return falls at or below the VaR quantile.

2.4 Holding-Period Scaling

Multi-day estimates are obtained via the square-root-of-time rule:

$$\widehat{\text{VaR}}_{\alpha}^{(H)} = \widehat{\text{VaR}}_{\alpha}^{(1)} \cdot \sqrt{H} \quad \widehat{\text{CVaR}}_{\alpha}^{(H)} = \widehat{\text{CVaR}}_{\alpha}^{(1)} \cdot \sqrt{H} \quad (4)$$

3 Parameters

Parameter	Type	Default	Description
window	int	250	Lookback window (number of trading days)
qtls	list	[0.05, 0.01, 0.001]	Tail probability levels
holding_period	int	1	Forecast horizon in days
decay	float	0.94	Defined but not used in HS (reserved)
verbose	bool	True	Print model setup information

Note: The `decay` parameter is accepted for interface consistency but has no effect in the Historical Simulation engine.

4 Advantages and Limitations

Advantages	Limitations
No distributional assumptions	Does not capture volatility clustering
Simple and transparent	VaR at 99.9% requires > 1000 observations
Captures fat tails empirically	Backward-looking; slow to adapt to new regimes
No calibration required	Square-root scaling assumes i.i.d. returns

5 Usage Example

```
from risk_analytics import risk_engines as re

var_hs = re.portfolio_var(
    rets_orig = rets,
    wgts       = weights,
    engine     = 'hs',
    fmt_engine = {
        'window'       : 500,
        'qtls'          : [0.05, 0.01, 0.001],
        'holding_period': 1,
    }
)
```